# CIT-315 Artificial Intelligence

## Chapter-1

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### The Turing Test approach:

The computer would need to posses the following capabilities:

1. Natural Language Processing (NLP)
2. Knowledge Representation (KR)
3. Automated Reasoning (AR)
4. Machine learning (ML)

To pass the total turing test, computer will need:

1. Computer vision
2. Robotics

These six disciplines compose most of AI.

### Defination of AI:

The four categories to define AI:

1. Acting Humanly – The Turing Test approach
   - AI as machines that behave like humans.
   - Passing the Turing Test (NLP, KR, AR, ML).
2. Thinking Humanly – The Cognitive Modeling approach
   - Focuses on understanding how humans think and replicating that process in machines.
   - Requires studying cognitive science and psychology.
3. Thinking Rationally – The 'Law of Thought' approach
   - AI as systems that follow logical reasoning (LoT).
   - Inspired by philosophers like Aristotle.

4. Acting Rationally – The Rational Agent approach.
   - Defines AI as rational agents that acts so as to achieve the best outcome or the best expected outcome.
   - Practical because it doesn't require mimicking humans, just rational agents.

## The Value Alignment Problem:
The problem of achieving agreement between our true preferences and the objectives we put into the machine is called the value alignment problem.
The values or objectives put into the machine must be aligned with those of the human.

## The foundations of AI:

1. Philosophy: Logic, reasoning, mind, action
2. Mathematics: Formal tools like- logic, probability, algorithms to describe reasoning and computation.
3. Economics: Introduced the idea of rational agents-decision makers who try to maximize outcomes.
4. Neuroscience: Studies how the brain enables intelligence, takes brain as inspiration for neural networks.
5. Psychology & Cognitive Science: Psychology studies how humans think, learn and solve problems.
6. Computer Engineering: The hardware/software that enables AI.
7. Control Theory & Cybernatics: Focuses on feedback systems, and goal-driven behavior.
8. Linguistics: AI needed insights of the structure of language to build neural language systems.

## Examples of AI achievements:

1. Robotics in industry and research
2. Web seach engines using AI techniques
3. Autonomous planning and scheduling
4. Machine translation
5. Speech recognition
6. Game-playing programs beating humans.

## Risks and Benefits of AI:

~~Risk~~
**Benefits:**

1. Automating repetitive tasks
2. Boosts productivity and innovation
3. Improving decision-making in medicine, logistics. ~~science~~.
4. Accelerates scientific discoveries (e.g., cures, climate change solutions)
5. Enhancing human capabilities (assistive technologies, smarter tools).

**Risks:**

1. Autonomous weapons could be misused.
2. AI may enable mass monitoring and manipulation.
3. AI can reflect and amplify social biases.
4. Automation could displace many workers.
5. AI failures in cars and healthcare can be fatal.
6. Future superintelligent AI could surpass human control if not aligned with human values.